# Divit Rawal

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**Research Interests.** Statistical learning theory (generalization bounds, sample complexity, decision theory); deep learning theory (training dynamics, implicit bias in overparameterized models, physics of learning); geometric statistics (information geometry, manifold priors).

## Education

**University of California, Berkeley** 2023 – 2027
*B.A. Statistics, Physics, Computer Science* *Berkeley, CA*

**Relevant Coursework:** Statistical Learning Theory†; Theoretical Statistics†; Mathematical Statistics†; Convex Optimization; Learning for Dynamics and Control†; Probability and Random Processes; Statistical Physics; Randomized Linear Algebra, Optimization, and Large-Scale Learning†. († denotes graduate-level)

## Publications and Preprints

[1] **A Theory of Saddle Escape in Deep Nonlinear Networks** (2026)
**Divit Rawal**, Michael R. DeWeese.
*Under review, NeurIPS.* [arXiv]

[2] **Rao-Blackwellized Score Matching on Manifolds** (2026)
**Divit Rawal**.
*ICML, Workshop on Structured Probabilistic Inference and Generative Modeling.* [arXiv]

[3] **Minimax Rates for Hyperbolic Hierarchical Learning** (2026)
**Divit Rawal**, Sriram Vishwanath.
*Preprint.* [arXiv]

[4] **ALPHANSO: Open-Source Modeling of $(\alpha, \mathbf{n})$ Neutron Source Terms** (2026)
**Divit Rawal**, Anthony J. Nelson, William Zywiec, Daniel Siefman.
*Forthcoming, Nuclear Instruments and Methods in Physics Research, Section A. 2026 ANS Student Conference; 2026 INMM Annual Meeting.* [arXiv] / [GitHub]

## Awards and Grants

**Best Paper Award, American Nuclear Society Student Conference** Apr 2026

Best Paper in Mathematics, Computation, and AI Applications for "ALPHANSO: Open-Source Modeling of $(\alpha, \mathrm{n})$ Source Terms," selected from $\sim 250$ submissions.

**VESSL AI Academia Grant** Aug 2025

Awarded for independent research into embeddings and information geometry; sole undergraduate recipient among a cohort primarily comprising Ph.D. students and postdoctoral researchers.

## Research Experience

**Berkeley Artificial Intelligence Research (BAIR)** Sep 2024 – Present
*Researcher* (advisor: Michael DeWeese) *Berkeley, CA*

Authored paper deriving an exact layer-imbalance identity for deep nonlinear networks and a critical-depth escape law, with matching theory and simulation under both symmetric and He-normal initialization.

Analyzed in-context learning in kernel ridge regression via eigenlearning, deriving mode-wise error dynamics and conditions for generalization without representation change.

Connected ICL behavior to spectral bias in kernel methods (eigenvalue decay, target alignment), clarifying how contextual examples induce effective adaptation in fixed-feature models.

**Lawrence Livermore National Laboratory** Mar 2025 – Present
*Researcher, Nuclear Science and Security Consortium Fellow* (advisor: Daniel Siefman) *Livermore, CA*

- Developed ALPHANSO, an open-source framework for deterministic modeling of $(\alpha, \mathrm{n})$ neutron production using continuous slowing-down approximations and modern evaluated nuclear data.
- Formulated stochastic and deterministic transport models for neutron yield and spectral prediction, benchmarking against legacy codes and experimental datasets.
- Validated against experimental benchmarks; demonstrated improved predictive accuracy and extensibility relative to legacy SOURCES-based pipelines.

**UC Irvine, Department of Physics & Astronomy** — Feb 2022 – Jul 2023
*Researcher* (advisor: Daniel Whiteson) — *Irvine, CA*

- Developed deep learning models for high-energy particle collision analysis, achieving 92% accuracy (vs. 81% baseline) in particle identification.
- Simulated collisions using MadGraph, Pythia8, Delphes, and ROOT; implemented reconstruction algorithms achieving 2% mass prediction error.
- Contributed to papers on neural simulation based inference (arXiv:2412.01600 and arXiv:2412.01548), both published in *Reports on Progress in Physics*.

## Invited Talks

**ALPHANSO: Open-Source $(\alpha, \mathbf{n})$ Neutron Source Terms**

- Lawrence Livermore National Laboratory — Mar 2026
- Lawrence Berkeley National Laboratory — Mar 2026
- University of Tennessee, Knoxville — Jun 2026

## Teaching

**Communication Networks (EE 122), UC Berkeley** — Spring 2026
*Head Teaching Assistant*

- Developed lab curriculum, wrote and graded exams and homeworks, and led weekly recitation sections.

**Communication Networks (EE 122), UC Berkeley** — Spring 2025
*Reader*

- Graded exams and homeworks.

**PCB Engineering (EE 198), UC Berkeley** — Spring 2024, Fall 2025
*Course Staff*

- Helped grade labs, provided feedback on and graded final projects.

## Industry Experience

**Cisco Systems, Foundation AI** — Jun – Aug 2025
*Software Engineer Intern* — *San Francisco, CA*

- Contributed to post-training and evaluation of cybersecurity-specialized LLMs built on Llama 3.1 8B, including instruction tuning and preference alignment (RLHF).
- Supported open-weight releases Foundation-Sec-8B and Foundation-Sec-8B-Instruct; contributed to accompanying technical reports on data curation, training methodology, and benchmark evaluation.

**ExperienceFlow AI** — May – Sep 2024
*Machine Learning Engineer Intern* — *Remote*

- Modeled FSM transition dynamics using transformer, state-space, and graph architectures; evaluated generalization and sample efficiency in low-label sequence prediction settings.
- Developed RL-based control policies (DQN, SARSA) for dynamic state reasoning systems, improving stability and convergence.

**Amazon** — Aug – Dec 2023
*Software Engineer Intern* — *Remote*

Implemented K-means clustering module in Java for OpenSearch ml-commons; increased unit test coverage from 66% to 78%.

Diagnosed and resolved production pipeline failures affecting 1M+ users, improving system reliability and reducing query latency.

## Technical Skills

**Programming:** Python, R, C++, Julia, Java, MATLAB, SQL, Mathematica, LaTeX.
**Scientific Computing:** NumPy, SciPy, Monte Carlo methods, randomized algorithms.
**ML/AI:** PyTorch, JAX, TensorFlow.